Grade 8 Name____________________
Unit 4 Date____________________
Lesson 3 Practice Period__________________

Apply the laws of exponents to write equivalent numerical expressions with one base and one positive exponent.

1. $\frac{5^{-5}}{5^{-7}} \cdot 5^4$

2. $\frac{((-2)^4)^3}{(-2)^2 \cdot (-2)^{-2}}$

3. $\frac{(1.3^4)^3}{(1.3^2)^2}$

4. $\left(\left(\frac{1}{3}\right)^0 \cdot \left(\frac{1}{3}\right)^{-3}\right)^2$

5. $\left(\frac{(4)^2 \cdot (4)^3}{(4)^{-2}}\right)^{-2}$

Consider the expression $\frac{\left(\frac{1}{2}\right)^3}{2^{-6}} \cdot 2^6$. For each expression in questions 6-10, determine if it is equivalent to $\frac{\left(\frac{1}{2}\right)^3}{2^{-6}} \cdot 2^6$. Then, justify your choice by explaining using laws of exponents or showing your work.

Expression	**Is It Equivalent to $\frac{\left(\frac{1}{2}\right)^3}{2^{-6}} \cdot 2^6$?**	**Justify or Explain**
6. $\frac{\left(\frac{1}{2}\right)^3}{\left(\frac{1}{2}\right)^6} \cdot 2^6$		
7. 2^{15}		
8. $\frac{(2)^3}{(2)^6} \cdot 2^6$		
9. 2^9		
10. $\frac{(2)^{-3}}{(2)^{-6}} \cdot 2^6$		